

History of Computers - Slideshow Information

Slide 1: What is a Computer?

- A computer is an electronic device that processes information by following instructions from hardware and software.
- Computers take input, process it, and produce output.
- Early computers were mechanical devices designed for calculations; modern computers are powerful digital machines.
- Fun Fact: The first computers were so large they could fill an entire room, yet they had less processing power than a modern smartwatch.

Slide 2: Who Invented the First Computer?

- Early mechanical calculators like the Pascaline (by Blaise Pascal in 1642) and Leibniz Calculator (by Gottfried Wilhelm Leibniz in 1673) set the groundwork.
- The Analytical Engine, designed by Charles Babbage in 1837, is considered the first conceptual computer.
- Babbage's Analytical Engine contained components similar to modern computers: an Arithmetic Logic Unit (ALU), control flow, and memory.
- Fun Fact: The Analytical Engine never made it past the blueprint stage, but modern computer scientists have since built working models based on Babbage's designs — and they worked!

Slide 3: How Did Computers Come to Be?

- Early devices like the Jacquard Loom (1801) introduced punch cards, an early form of programming.
- In 1890, Herman Hollerith developed a Tabulating Machine that used punch cards to quickly calculate census data
- WWII computing advances, like the Colossus (1944) and ENIAC (1945), dramatically improved data processing
- Fun Fact: The Colossus was classified top-secret for decades — even its engineers couldn't talk about their groundbreaking work until long after WWII ended.

Slide 4: Who is Charles Babbage?

- Known as the "Father of the Computer," Babbage designed the Analytical Engine in 1837.

His design included:

- A mill (the CPU) for calculations
- A store (memory) for data storage
- Input and Output devices for receiving data and displaying results
- Fun Fact: Babbage was known for his eccentric personality. He once attempted to ban street musicians because their noise distracted him from his work!

Slide 5: Who Was Ada Lovelace?

- Ada Lovelace was born in 1815, the daughter of poet Lord Byron and mathematician Annabella Byron. Her mother pushed her toward math and logic, believing it would keep her from inheriting her father's "wild imagination." Ironically, Ada blended both creativity and logic in her work.
- Ada became intrigued by Charles Babbage's Analytical Engine. In 1843, she translated an article about the machine and added her own detailed notes — these notes included what's now considered the first computer algorithm.
- Ada predicted that computers could one day create music, art, and complex designs — a bold vision that closely resembles how we use technology today.
- Ada is known as the "first computer programmer", and her contributions were overlooked until the 1950s. Today, the Ada programming language and Ada Lovelace Day celebrate her legacy.
- Fun Fact: Ada once described her work as "poetical science", blending imagination with logic to predict the potential of computers far beyond her time.

Slide 6: Who Was Alan Turing?

- Alan Turing (born in 1912) was a British mathematician, cryptographer, and the "Father of Modern Computing." Turing was known for his brilliance, and was fascinated by puzzles at a young age.
- During World War II, Turing worked at Bletchley Park, where he designed the Bombe Machine, which

cracked the German Enigma Code. Historians estimate Turing's code-breaking work shortened the war by two years and saved 14 million lives.

- Turing's biggest breakthrough came in 1936 when he proposed the Turing Machine, a theoretical device that laid the groundwork for modern computers. His concept of a machine capable of following instructions for any task mirrors how modern computers operate today.
- Turing was also one of the first to ask: "Can machines think?", inspiring what is now known as the Turing Test, a measure of machine intelligence.
- Fun Fact: Turing was a talented long-distance runner and nearly qualified for the 1948 Olympic Games. He often ran 40 miles just to attend team meetings at Bletchley Park!

Slide 7: Who Was Grace Hopper?

- Grace Hopper (born in 1906) was a computer scientist and U.S. Navy Rear Admiral who revolutionized programming. Known as "Amazing Grace," Hopper was a trailblazer in both tech and the military.
- In 1952, Hopper developed the first compiler, a program that translates written instructions into machine code. This innovation made programming accessible to non-experts.
- Hopper played a major role in developing COBOL, one of the earliest and most widely used programming languages for business.
- In 1947, Hopper famously coined the term "debugging" after her team found a moth inside the Harvard Mark II computer — the first recorded computer bug.
- Fun Fact: Grace Hopper carried a piece of nanosecond wire (about 30 centimeters long) to demonstrate how far electricity travels in one-billionth of a second — showing engineers why faster computing required shorter circuits.

Slide 8: How Did We Go from Mechanical to Digital?

- Early computers like the Analytical Engine were mechanical, using gears and levers to calculate data.
- The transition to electronic and digital computers

began in the 1940s with machines like Colossus (1944) and ENIAC (1945).

- Vacuum tubes replaced mechanical parts, allowing data to be processed faster and more efficiently.
- By the 1950s, computers moved from vacuum tubes to transistors, dramatically reducing size and power consumption.
- In the 1970s, the development of the microprocessor allowed for small, personal computers that revolutionized modern computing.
- Fun Fact: Early computers were so large that some filled entire rooms, yet they were less powerful than a modern smartphone!

Slide 9: Modern Computing - First Personal Computer

- The first widely recognized personal computer was the Altair 8800 (1975).
- The Apple I (1976) and IBM PC (1981) made computing accessible to homes and businesses.
- The Macintosh (1984) introduced a graphical user interface (GUI), making computers easier for non-technical users.
- Fun Fact: The first Apple computer was sold as a kit — buyers had to assemble it themselves!

Slide 10: Silicon Valley's Role in Computing History

- Silicon Valley is the birthplace of major computing innovations.
- Hewlett-Packard (HP), founded in 1939, created precision instruments and produced one of the first programmable desktop computers — the HP 9100A (1968).
- Fairchild Semiconductor (1957) developed key technology that led to the microchip revolution.
- Intel (1968) created the Intel 4004, the world's first commercial microprocessor.
- Apple (1976) revolutionized personal computing with the Apple I and Macintosh.
- Modern tech giants like Google, Facebook, and Netflix emerged from Silicon Valley, continuing its tradition of innovation.
- Fun Fact: The first Google server was built using Legos!

Citations on the Web for the Slideshow

Slide 1: What is a Computer?

Source: Computer History Museum, "Timeline of Computer History"

<https://www.computerhistory.org/timeline/computers/>

Slide 2: Who Invented the First Computer?

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Slide 3: How Did Computers Come to Be?

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Slide 4: Who is Charles Babbage?

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Slide 5: Who Was Ada Lovelace?

Source: The Lifewire article, "Meet the Women Who Revolutionized Technology"

<https://www.lifewire.com/women-in-tech-11689509>

Slide 6: Who Was Alan Turing?

Source: The Times, "The Times view on Alan Turing: Decoding Genius"

<https://www.thetimes.co.uk/article/the-times-view-on-alan-turing-decoding-genius-5tbd50r5c>

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Source: Lifewire, "Meet the Women Who Revolutionized Technology"

<https://www.lifewire.com/women-in-tech-11689509>

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Presentation Guidelines

Objective: Create a short presentation about a computing pioneer or invention from the class slideshow.

Step 1: Drawing Your Topic

- Each student (or group) will randomly draw their presentation topic from one of the slides in the computer history presentation.
- Topics may include inventors like Ada Lovelace, Alan Turing, or Grace Hopper, or inventions like ENIAC, the Turing Machine, or the Apple I.
- Once you've drawn your topic, you cannot switch — embrace the challenge!

Step 2: Presentation Format

You may choose one of the following formats for your presentation:

- **The Time Traveler's Report** – A TEDx Talk-style speech where students “travel back in time” to explain how the invention will change the world.
- **The Tech Detective** – A true-crime style video investigating the social impact of an invention.
- **The Digital Museum Exhibit** – A visual presentation showcasing the invention's features, history, and influence
- **The Visual Timeline Project** – A visual timeline connecting the invention to modern technology.

Step 3: Presentation Length

- Presentations must be between 1 minute 30 seconds and 3 minutes 30 seconds.
- Time management is important — practice your presentation to ensure it fits within this range.

Step 4: What to Include in Your Presentation

Your presentation should answer the following questions:

1. Who was the inventor or what is the invention?
2. When was this invention created or when did the inventor live?
3. What problem did this invention solve, or what contribution did the inventor make to computing?
4. Why is this invention or inventor important today?

5. Share one interesting fact or memorable detail about your topic.
6. Bonus: Creative elements such as visuals, props, or storytelling are encouraged.

Step 5: Presentation Tools

You may choose one of the following methods to present your information:

- Google Slides / PowerPoint
- Poster or Visual Board
- Skits, Role-Play, or Creative Storytelling
- Short Video Presentation (For students who prefer pre-recording)

Step 6: Grading & Expectations

Your presentation will be assessed based on:

- Clarity & Organization (Is your information clear and structured?)
- Content Accuracy (Did you include key facts about your topic?)
- Creative Elements (Did you use visuals or creative storytelling?)
- Collaboration (If working with a partner or group, did everyone participate?)
- Timing (Staying within the time limit)

Step 7: Practice & Prepare

- Practice your presentation to improve your delivery and timing.
- Remember: Speaking clearly, using eye contact, and showing confidence will make your presentation stronger.

Final Note:

Have fun! This is your chance to creatively share the story of a computing pioneer or invention that shaped the world we live in today.

Name(s): _____

Topic Drawn: _____

Presentation Format: _____

Step 1: Key Information

What is your topic?

(Describe the invention or inventor in one sentence)

When was this invention created, or when did the inventor live?

What problem did this invention solve, or what contribution did the inventor make to computing?

Why is this invention or inventor important today?

One interesting fact or memorable detail about your topic:

Step 2: Presentation Format

Which presentation format are you using? (Check one)

- ☐ **The Time Traveler's Report** – A TEDx Talk-style speech where students “travel back in time” to explain how the invention will change the world.
- ☐ **The Tech Detective** – A true-crime style video investigating the social impact of an invention.
- ☐ **The Digital Museum Exhibit** – A visual presentation showcasing the invention's features, history, and influence.
- ☐ **The Visual Timeline Project** – A visual timeline connecting the invention to modern technology.

Step 3: Presentation Plan

Opening Idea:

(How will you introduce your topic?)

Main Points to Cover:

- 1.
- 2.
- 3.

Conclusion Idea:

(How will you wrap up your presentation?)

Step 4: Creative Ideas

What visuals, props, or creative elements will you include?

Who is responsible for each part of the project?

Step 5: Checklist for Success

- ☐ My presentation answers all 5 key questions.
- ☐ My presentation fits the time limit (1:30 – 3:30).
- ☐ I have included at least one creative element.
- ☐ I have practiced my presentation.
- ☐ I am prepared to confidently present my project.

Name: _____

Topic: _____

Step 1: Reflecting on Your Presentation

Think about your presentation and answer the following questions honestly.

1. What went well in your presentation?
2. What was challenging for you during this project?
3. If you could improve one part of your presentation, what would it be?
4. How did your group work together (if applicable)?
5. What is one thing you learned about your topic that surprised you?

Step 2: Rubric Self-Assessment

Use the Provided Rubric and grade yourself on each section.

Why do you feel you scored that way? Did you learn any new skill during this presentation?

	4 (Excellent)	3 (Good)	2 (Fair)	1 (Poor)
Computer Literacy / Communication	Clear, organized presentation with accurate information. The invention's purpose, function, and impact are explained effectively.	Presentation is mostly clear with minor gaps in explanation or organization.	Presentation is somewhat organized but lacks clear explanations or details.	Presentation is unclear, disorganized, or missing key information.
Impacts of Computing	Effectively explains how the invention influenced society, technology, or ethics. Strong connections to modern computing.	Explains some societal or technological impacts with minor gaps in detail.	Provides minimal explanation of the invention's impact or relevance.	No clear explanation of the invention's impact or connections to modern computing.
Design	Visuals are clear, organized, and enhance the presentation. Strong creative elements support understanding.	Visuals are present and somewhat effective, but may lack creativity or clarity.	Visuals are present but poorly organized or ineffective.	No visuals included, or visuals are confusing and unhelpful.
Collaboration	All team members contributed meaningfully. Strong evidence of teamwork and shared responsibility.	Most team members contributed equally with minor imbalances.	One or two students did most of the work; collaboration was limited.	Little evidence of collaboration; work appears disjointed.